

have ample capital and top talent; however, their corporate DNA and ingrained processes limit their agility. In contrast, software demands rapid iterative updates and flexibility—traits that are native to startups like ours. This agility enabled autonomous driving systems to emerge from startups. While large automakers possess the necessary capabilities, they often make strategic investments in startups to innovate without disrupting their own operational DNA. This approach enables them to adapt to the software-defined vehicle transition and other emerging technologies like AI without deviating from their primary focus on automotive challenges. Our success stems from leveraging our nimble, adaptive nature to solve problems that require rapid, innovative solutions.

To address the first question, while we develop the AI models, this opens significant opportunities for electronics around sensors, such as cameras, and edge computing to deploy these models in vehicles.

Throughout your journey, you have partnered with major companies widely recognised in the media. Have smaller, lesser-known companies also played a role in your projects?

Absolutely. Software cannot function without hardware, underscoring the diverse contributions to our projects. For example, we use various components, ranging from small inertial measurement units available from major companies like Bosch, Sony, and Nvidia, to smaller specialists such as Xsens and e-con Systems. Emerging startups like Mindgrove from IIT Madras, which recently developed India's first fabless silicon, excel in this area. Additionally, our projects often integrate specific solutions like compact networking units to support telematic systems. Our company operates as one of many tier-one providers within a broader ecosystem that includes numerous tier-two suppliers

and collaborators. This extensive network is crucial for delivering high-performance, optimised products in technologically advanced fields like automotive development.

Do India's electronics and hardware capabilities support local software and algorithm production, or must these components be sourced internationally?

India lacks the electronic capabilities to compete with major global players in the silicon industry. While India can produce basic components like transistors, more complex and essential parts are predominantly sourced from Taiwan, often via the US or Singapore. This includes products from major companies such as Qualcomm, Intel, and Nvidia. The Indian market operates on a much smaller scale, focusing on a limited segment of the electronics industry. For instance, in the area of camera sensors, dominated by Sony and Onsemi, the related infrastructure and ISPs are often developed by a few international OEMs like Valeo and Bosch, with only a handful of Indian startups like e-con Systems contributing. However, there is still a significant gap in silicon. While there are many other startups like Sima.ai building AI silicon and doing their engineering in India, they remain US-headquartered companies.

Have you explored potential partnerships with Indian companies, and if not, why do you think collaboration is lacking?

Yes, we have explored local collaborations. It is important to recognise that many Indian companies are actively developing technologies at the software layer, and some are working with smaller-scale hardware like FPGA cards. At this year's expo, several Indian companies will showcase processors in the 5-watt range based on ARM, which are suitable for applications like automotive

telematics systems that we can source entirely within India.

However, challenges arise when dealing with higher power requirements—above 50 or 60 watts—where the hardware often centres around CPUs or neural processing units. These are typically provided by leading global firms like Nvidia and Qualcomm, which also supply the reference designs. While there are many competent tier-two silicon suppliers in India, they must still rely on partnerships with these major tier-one hardware companies to assemble complete systems. Thus, while there are opportunities at certain levels, integration with top-tier hardware providers remains limited in India.

Are you suggesting that in India only less complex electronics with lesser value addition are available, whereas more valuable hardware components are not?

Yes, exactly. In India, we have capabilities for lower-power applications like telematics systems and sensors, which operate below 5 watts. These sectors benefit from technologies developed locally about 20 years ago, with much of the intellectual property now being open. However, the situation is different for more advanced silicon technologies used in higher-power electronics. The fabrication designs and related patents, which are crucial for this segment, remain tightly controlled globally and are only about 15 to 16 years old—still too young to be open to the Indian market. Although we see promising developments from companies like Mindgrove, significant local advancements in high-value hardware may still take a couple of years. I remain optimistic about the future, but at present, India lacks the capability to produce these high-value components independently. Nevertheless, I see this as a positive sign, as it highlights a huge opportunity for future development. **EFY**